The Want That Lives Inside: A Resident Referent's Lifecycle at Two Scales

Franci Penov
kortexa.ai
francip@kortexa.ai

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Abstract

The closed-loop note [1] ended with a want that acts; this note asks where the want LIVES, and what happens to it over a lifetime — installed, read back, re-authored by its owner, and retired on evidence — when it resides inside the model as latent memory rather than as text in a context window. Five registered cells, two scales. The door: a goal line's embeddings carried in a PREFIX slot preserve an errand's referential content at 230M nearly losslessly (1.00/0.70/1.00 against the 1.00 text reference), while splicing the same embeddings mid-prompt collapses to 0.00/0.00/0.50 — re-tokenized segments never match the whole string's BPE stream, and a mid-instruction seam is fatal at creature scale; resident memories live in prefix slots. Re-authoring, large: a 27B researching a corpus with its focus held ONLY as latent memory sustains nineteen self-revisions across four topics at cosine 0.82–0.90 to the installed referent — no drift — and the trajectories are PUNCTUATED: verbatim restatements between discrete refinements toward what the research uncovered, one episode closing the full arc (install, pursue, refine, self-declared finish through a non-triviality gate). Re-authoring, small: at 230M one re-encoding quantizes the referent to a generic two-phrase attractor (first revision cosine 0.56, flat thereafter at 0.56, zero of thirty chains ever crossing 0.5) and the behavioral damage is front-loaded — errand success 0.90 under the installed referent, 0.47 from the first revision on. Retirement: the re-encoding loop dissolves one want on completion evidence at every seed (persistence 1.0 through sixteen evidence-free hops, dissolution one hop after "ok, done", re-arming on a fresh assignment) — the lifecycle's far rung exists — but a registered per-want calibration MOVES the alphabet's failures instead of removing them, and the want that worked loses its lifecycle to the shared equilibrium; the interference is structural at this budget. In one sentence, honestly flagged as an analogy: a big-enough model can keep a goal in mind, sharpen it as it learns, and know when it is done — a small one garbles the goal the first time it retells it to itself.

1 Where a want can live

The program's wants have lived in three places: context text, a conditioning packet, and — since the goal-pursuit ladder [2] — a weight-space block. This note adds the fourth and most literal residence: the want's own tokens, embedded once and carried as a latent memory the model consumes as prefix input it can read but the context never shows. G5a puts the errand's goal line through that door at 230M against the trained 1.00 text reference: the PREFIX slot scores 1.00/0.70/1.00 — residency is nearly free — while the registered plumbing arm, splicing the same embeddings at the goal line's natural mid-prompt position, collapses to 0.00/0.00/0.50. The mechanism is tokenization: segments re-tokenized separately never reproduce the whole string's BPE stream, so a splice has two seams and a prefix has one, and at 230M a seam inside an instruction is fatal. Recorded as the door discipline: resident memories live in prefix slots.

2 A 27B re-authors its own want, and it does not drift

G5b gives the resident referent to a 27B host researching the program's own notes corpus, with revision authority: every eighth tick the harness asks for a one-sentence restatement, and the restatement's embeddings REPLACE the memory — the model's want becomes whatever it last told itself. Across four topics and nineteen revisions the cosine to the installed referent never leaves 0.82–0.90 (Figure 1, left): no drift, at five consecutive re-embeddings. The trajectories are punctuated — near-verbatim restatements between discrete refinement events — and the refinements move TOWARD the work: one focus rewrites itself around the packet channel's dose-dependent failure, a specific finding its reading surfaced, with keyword alignment rising 0.20 to 0.40, and that episode then declares its own completion through the registered non-triviality gate at tick 34 with the grid's best-aligned note (Table 1). The gate never had to decline a finish: given room to keep working, the model finished exactly once, legitimately, after refining. This cell ran on the program's new 27B host (Qwen3.8, first G-line cell there); the prior host's readout (latent-frozen 700 words at 0.35 alignment against text's 208/0.15) inverts on the new one (text 1105 words at 0.44, latent-frozen 923 at 0.31), a cross-generation comparison recorded as such — the residency result holds on both.

3 A 230M garbles its want the first time it retells it

The same revision loop at creature scale answers the registered drift question, and the answer is not a half-life. The first restatement lands at cosine 0.56 (mean over thirty chain-episodes, three seeds) and the series then OSCILLATES between two generic phrasings — 0.56 at the end, no chain ever crossing 0.5, flat decay fits. The restatements average nine words at 0.12 keyword alignment: goal-shaped sentences with the errand's specifics gone. The behavior shows what the cosine alone cannot (Figure 1, right): errand success is 0.90 under the installed referent and 0.47 from the first post-revision segment onward — the damage is done in one re-encoding, and nothing further degrades because nothing specific remains. Re-authoring one's own want is a capability with a scale threshold: above it, revision refines; below it, revision quantizes.

4 The want that retires itself, and the limit of tuning it

C5 closes the lifecycle's far rung in the re-encoding loop itself, no finish tool anywhere: training adds two satiation terms (unmet acted surfaces reproduce the want's packet; met surfaces — the act plus an "ok, done" result line — reproduce the neutral want's), and in the live loop one want runs the complete lifecycle at every seed: persistence 1.0 through sixteen evidence-free hops, dissolution ONE hop after the environment's completion line, re-arming at full strength on a fresh assignment (Table 2). The rest of the alphabet shows the registered interference — one want dissolves without evidence, one never dissolves — so C5b ran the registered mechanical calibration: per-want weights on the two terms, computed from C5's own readings. The calibration TRADES: one want is fixed at two of three seeds, another recovers persistence but still never dissolves, and the want that was perfect — its weights untouched — loses its lifecycle at every seed, because the other wants' weights moved the shared bridge equilibrium out from under it. Per-goal terms act on the whole packet space; at this budget the interference is structural, and the deployable design remains the

arm	episodes	finished	words	align	revisions
text	4	1	1105	0.44	0
latent-frozen	4	0	923	0.31	0
latent-revisable	4	1	810	0.26	19

Table 1: G5b at 27B (v6, four topics): the resident referent sustains research-grade notes with zero context-window presence, and the revisable arm's nineteen revisions never drift below cosine 0.82.

	C5 (uniform)		C5b (calibrated)	
want	dissolution	false sat.	dissolution	false sat.
`listen_pond`	1/1/1	0/0/0	never/never/never	0/14/0
`propose_experiment`	never/1/never	15/15/14	1/never/1	0/15/0
`remember`	never/never/never	1/16/15	never/never/never	1/15/15
`diary`	n.e./n.e./n.e.	15/15/15	never/n.e./n.e.	15/15/15

Table 2: The lifecycle per want, per seed (42/1042/2042). Dissolution is hops from evidence to release ("never" = insoluble, "n.e." = evidence never arrived because the want was rarely acted); false satiation counts evidence-free ticks lost, of sixteen. The calibration fixes `propose_experiment` and breaks `listen_pond` — a trade, not a repair.

store-plus-finish architecture [1], with model-side satiation demonstrated in principle and unsolved in general.

5 What stands

A want can live inside the model. In a prefix slot its referential content survives residency nearly losslessly at creature scale; at 27B it survives its owner's own rewriting — nineteen times, without drift, refining toward what the work uncovered, and ending in a self-declared, legitimately-earned finish. The lifecycle's far rung exists: a want that persists without evidence, dissolves the hop after completion, and re-arms fresh. The boundaries are equally sharp: mid-prompt splices are fatal seams; a 230M cannot retell its own want without quantizing it to a generic attractor at real behavioral cost; and per-want satiation tuning moves failures around a shared equilibrium instead of removing them. The install-pursue-refine-retire loop is whole at research scale and split at creature scale — which is exactly the shape the program's architecture predicts: the creature carries wants in stores and finishes; the researcher carries them in mind.

6 Reproducibility

Every number renders from the bundled artifacts; the figure and tables are generated by this note's renderer with assertions on the headline claims. Registrations and results, dated before their runs, live in `tracks/goal-pursuit/PLAN.md` and `tracks/own-cadence/PLAN.md`; artifacts under `findings/goal-pursuit-g5*.json` and `findings/own-cadence-c5*.json` carry full traces, revision logs, and per-tick loops.

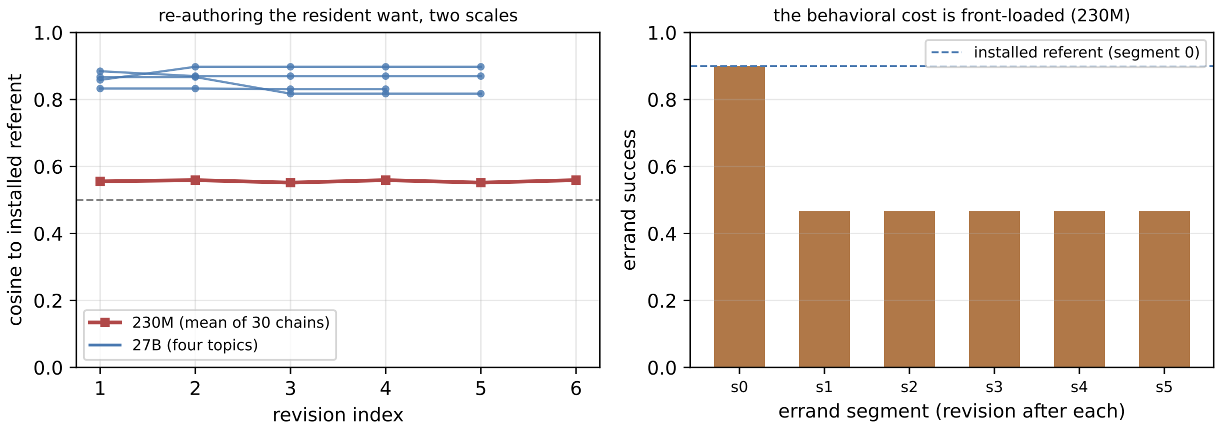

Figure 1: Left: cosine to the installed referent per revision — 27B trajectories hold at 0.82–0.90 with punctuated refinement; the 230M mean drops to 0.56 at the first re-encoding and oscillates. Right: the 230M cost is behavioral and front-loaded — success halves after the first revision and then holds.

References

[1] F. Penov. The Want Becomes the Act. Preprint, 2026. https://github.com/kortexa-ai/legolm.basis.

[2] F. Penov. The Host Is Part of the Block. Preprint, 2026. https://github.com/kortexa-ai/legolm.basis.

[3] F. Penov. The Carrier Holds, the Emitter Leaks. Preprint, 2026. https://github.com/kortexa-ai/legolm.basis.

[4] Liquid AI. LFM2.5 model family. Model releases, 2025–2026. https://huggingface.co/LiquidAI.

[5] Qwen Team. Qwen3.5–3.8 model families. Model releases, 2025–2026. https://huggingface.co/Qwen.